K/U	A	T	C	Total
20	8	10	10	48

Short Answer (T/C) – 2 marks each

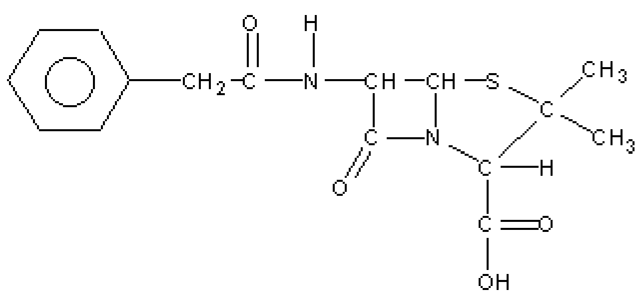
1. Benzoic acid is used as a preservative in maraschino cherries. High levels are added to prevent spoilage. Sketch the structure of benzoic acid and comment on why its inclusion in food should be a concern for consumers.

2. In household fires, it is often the smoke produced that creates a danger as great as the flames. Many plastic materials in the house contain polyvinylchloride (PVC); the formula for this compound is $(C_2H_3Cl)_n$. Write an equation for the combustion of PVC in oxygen. Suggest a reason why its incineration poses a serious risk during a fire.

3. Propane can be used as starting material for either propanal or propanone. Explain how this is possible.

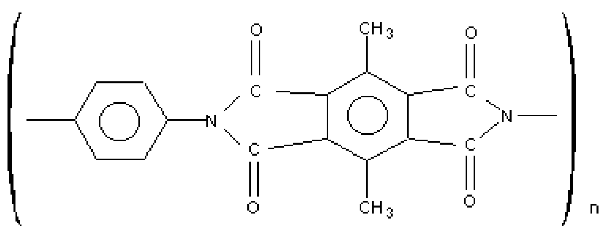
4. 2-butanol is oxidized. Suggest a method a chemist could use to separate the oxidation product from the original 2-butanol. Include the condensed formula of the oxidation product in your answer.

5. The structure given below is PenicillinG, an antibiotic. Circle the functional groups in the molecule and list their corresponding names.



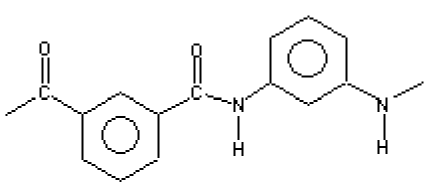
6. During a school laboratory test, you are given 2-pentyne and asked to prepare 2-pentanone and 3-pentanone from it. Develop a synthesis pathway to accomplish this task.

7. The structure given below is a polymer classified as a polyamide. This polymer is stable in temperatures above 350°C, and can be used as a protective coating on hot surfaces. What monomers were used to make this polyamide?

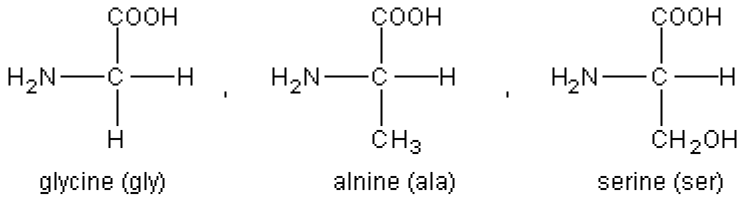


8. Formaldehyde (methanal: $H_2C{=}O$) can be polymerized to produce Delrin (polyoxymethylene), a strong and easily molded plastic. Write a chemical reaction that shows the formation of this polymer.

9. Nomex is a polymer used to make flame-resistant clothing for firefighters. A portion of its structure is provided below. Write a polymerization reaction showing its production from monomers. What type of reaction is this?



10. The structures of three amino acids are given below. Draw and label 4 simple 2 amino acid polypeptide structures which can be produced from these amino acids (example of structure label: gly-ala).



Essay (MC) Answer only ONE of the following questions. (8 marks)
Provide sufficient detail and supporting points. Answer in an organized paragraph.

11. Designing a new pharmaceutical drug to treat an illness or chronic condition is an enormous challenge. Describe some of the problems encountered in the manufacture and utilization of a new medicine. Discuss thalidomide and explain the issue of chirality (stereoisomers or enantiomers).

12. Polymers are often thought to be manufactured in immense chemical plants. Although many are, there is a large group of polymers which are natural (biopolymers). Describe several natural polymers and how they are utilized in our society or in our bodies.